

Reproducibility Checklist

This checklist is filled for the current BGR submission package. The main paper is anonymous; commands, configs, per-seed CSVs, and run ledgers are in the anonymous artifact's `README.md` and `results/README.md`.

1. General Paper Structure

- 1.1. Includes a conceptual outline and/or pseudocode description of AI methods introduced (yes/partial/no/NA)
yes. The method section defines replayable states and recovery curves. It also states critical radii, BGR priority, and a training-loop method box.
- 1.2. Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results (yes/no)
yes. The paper separates controlled margin-expansion claims, robot-suffix simulator results, OpenVLA fixed-policy audits, OpenVLA-OFT adaptation diagnostics, and tabular grid-policy setup failures.
- 1.3. Provides well-marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper (yes/no)
yes. The related-work section cites curriculum learning, PLR, PAIRED/ACCEL, domain randomization, MiniGrid, LIBERO, and OpenVLA.

2. Theoretical Contributions

- 2.1. Does this paper make theoretical contributions? (yes/no)
yes. The paper introduces a recovery-margin formalism and states a local margin-update proposition for boundary-centered radius sampling.

If yes, please address the following points:

- 2.2. All assumptions and restrictions are stated clearly and formally (yes/partial/no)
yes. Proposition 1 states the local update-kernel and stochastic-dominance assumptions; the limitations section states replay-access and perturbation-family restrictions.
- 2.3. All novel claims are stated formally (e.g., in theorem statements) (yes/partial/no)
yes. The local theoretical claim is stated as Proposition 1. Empirical claims are stated with measured metrics and paired-seed tests.
- 2.4. Proofs of all novel claims are included (yes/partial/no)
yes. Proposition 1 includes a proof sketch based on monotone coupling and monotonicity of the update kernel.
- 2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no)
yes. The method section gives the local margin-update intuition and the proposition proof sketch.
- 2.6. Appropriate citations to theoretical tools used are given (yes/partial/no)
yes. The proof cites a standard stochastic-orders reference for the quantile coupling used in Proposition 1.
- 2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA)
partial. The proposition is a local model; the grid ablation empirically tests its main implication that boundary-centered radius sampling matters.
- 2.8. All experimental code used to eliminate or disprove claims is included (yes/no/NA)
yes. Ablation scripts, configs, per-seed CSVs, and claim-checking scripts are included.

3. Dataset Usage

- 3.1. Does this paper rely on one or more datasets? (yes/no)
partial. Main BGR training results use procedural synthetic/grid/suffix generators. The LIBERO/OpenVLA recovery audit

uses existing benchmark task definitions, init states, and fixed-policy rollout artifacts to validate learned-policy recovery-curve measurement.

If yes, please address the following points:

- 3.2. A motivation is given for why the experiments are conducted on the selected datasets (yes/partial/no/NA)
yes. The paper motivates synthetic recovery curves, procedural grid-margin replay states, robot suffix diagnostics, and LIBERO/OpenVLA as a realistic manipulation-policy target.
- 3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA)
yes. The experiments are generated by artifact-provided scripts/configs; per-seed result CSVs are included under results/.
- 3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA)
yes. The generated procedural data and code are included in the anonymous artifact under an MIT license.
- 3.5. All datasets drawn from the existing literature (potentially including authors' own previously published work) are accompanied by appropriate citations (yes/no/NA)
yes. LIBERO and OpenVLA are cited; MiniGrid is cited for grid-style context.
- 3.6. All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available (yes/partial/no/NA)
yes for LIBERO task definitions, benchmark assets, and OpenVLA model code/weights. The anonymous artifact contains compact derived recovery summaries; raw fixed-policy rollout logs are retained for audit but are not required for the claimed quantitative results.
- 3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisfying (yes/partial/no/NA)
NA. No non-public dataset is used for claimed quantitative results.

4. Computational Experiments

- 4.1. Does this paper include computational experiments? (yes/no)

yes.

If yes, please address the following points:

- 4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA)
partial. Final configs and ablation variants are included in the anonymous artifact; the full search history is not exhaustively enumerated in the main paper.
- 4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no)
yes. Procedural generation, LIBERO probing, OpenVLA recovery summarization, aggregation, and figure scripts are included.
- 4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no)
yes. Experiment scripts, configs, tests, aggregation, and result CSVs are included in the anonymous artifact.
- 4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no)
yes. The anonymous artifact includes an MIT license, and the code plus generated procedural data will be made publicly available upon publication.
- 4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no)
partial. Code is modular and tested; comments are intentionally sparse and do not yet cross-reference paper sections.

- 4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results (yes/partial/no/NA)
yes. Every config lists seeds; scripts pass seeds to NumPy generators and result CSVs include per-seed rows.
- 4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no)
yes. The run ledger records Slurm partitions, CPU/memory/time requests, hostnames for logged runs, environment constraints, and JSON environment snapshots for compute and GPU jobs.
- 4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no)
yes. Recovery AUC, critical radius, clean success, transfer RAUC, and AULC are defined or described.
- 4.10. This paper states the number of algorithm runs used to compute each reported result (yes/no)
yes. Runs are reported in the paper and run ledger: synthetic and grid learning-curve diagnostics use 15 seed pairs; estimator validation, grid-margin full-baseline, grid ablation, grid sensitivity confirmations, and robot-suffix coverage comparisons use 30 paired seeds. Smaller three- or five-seed exploratory diagnostics are labeled as diagnostics in the run ledger rather than used as main claims.
- 4.11. Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no)
yes. Aggregated tables include standard errors; experiments report median critical radius and AULC in addition to final means.
- 4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no)
yes. The anonymous artifact includes paired exact sign tests for the reported paired comparisons, including the 30-seed suffix coverage confirmation, grid-margin target-sensitivity, grid-margin obstacle-regime sensitivity, grid-margin geometry-stress sensitivity, and grid-margin learning-curve sweeps. The paper separates these main comparisons from smaller exploratory evidence.
- 4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA)
yes. Final hyperparameters are in the artifact YAML configs under `configs/`.